

torch.linalg.matrix_norm#

https://pytorch.org/docs/stable/generated/torch.linalg.matrix_norm.html

Description:

Computes a matrix norm. If A is complex valued, it computes the norm of `A.abs()`. Support input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices: the norm will be computed over the dimensions specified by the 2-tuple `dim` and the other dimensions will be treated as batch dimensions. The output will have the same batch dimensions. `ord` defines the matrix norm that is computed. The following norms are supported:

Function Signature

```
torch.linalg.matrix_norm(A, ord='fro', dim=(-2, -1),  
keepdim=False, *, dtype=None, out=None) → Tensor#
```

Parameters

Keyword Arguments

Returns

Parameters

Keyword

out (Tensor, optional) – output tensor. Ignored if None. Default: None. dtype (torch.dtype, optional) – If specified, the input tensor is cast to dtype before performing the operation, and the returned tensor's type will be dtype. Default: None

Returns:

A real-valued tensor, even when A is complex.

Code Examples

```
>>> from torch import linalg as LA
>>> A = torch.arange(9, dtype=torch.float).reshape(3, 3)
>>> A
tensor([[0., 1., 2.],
        [3., 4., 5.],
        [6., 7., 8.]])
>>> LA.matrix_norm(A)
tensor(14.2829)
>>> LA.matrix_norm(A, ord=-1)
tensor(9.)
>>> B = A.expand(2, -1, -1)
>>> B
tensor([[[0., 1., 2.],
         [3., 4., 5.],
         [6., 7., 8.]],
        [[0., 1., 2.],
         [3., 4., 5.],
         [6., 7., 8.]]])
>>> LA.matrix_norm(B)
tensor([14.2829, 14.2829])
>>> LA.matrix_norm(B, dim=(0, 2))
tensor([ 3.1623, 10.0000, 17.2627])
```

Detailed Documentation

Documentation

Computes a matrix norm.

If A is complex valued, it computes the norm of `A.abs()`

Support input of float, double, cfloat and cdouble dtypes. Also supports batches of matrices: the norm will be computed over the dimensions specified by the 2-tuple `dim` and the other dimensions will be treated as batch dimensions. The output will have the same batch dimensions.

`ord` defines the matrix norm that is computed. The following norms are supported:

`max(sum(abs(x), dim=1))`

`min(sum(abs(x), dim=1))`

`max(sum(abs(x), dim=0))`

`min(sum(abs(x), dim=0))`

largest singular value

smallest singular value

where `inf` refers to `float('inf')`, NumPy's `inf` object, or any equivalent object.

`A` (Tensor) – tensor with two or more dimensions. By default its shape is interpreted as `(*, m, n)` where `*` is zero or more batch dimensions, but this behavior can be controlled using `dim`.

`ord` (int, inf, -inf, 'fro', 'nuc', optional) – order of norm. Default: 'fro'

`dim` (Tuple[int, int], optional) – dimensions over which to compute the norm. Default: (-2, -1)

`keepdim` (bool, optional) – If set to True, the reduced dimensions are retained in the

result as dimensions with size one. Default: False

out (Tensor, optional) – output tensor. Ignored if None. Default: None.

dtype (torch.dtype, optional) – If specified, the input tensor is cast to dtype before performing the operation, and the returned tensor's type will be dtype. Default: None

A real-valued tensor, even when A is complex.